

HAVRE, MT, USA

WMO: 727770

Lat: 48.543N

Lon: 109.763W

Elev: 2585

StdP: 13.37

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94012

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-14.8	-26.2	1.4	-19.9	-19.6	2.1	-13.1	32.0	39.0	28.6	37.3	6.2	230	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.1	95.0	63.6	91.4	62.8	87.8	61.9	67.3	86.0	65.3	84.7	63.6	82.7	10.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	89.0	73.1	58.9	81.8	70.1	56.7	75.4	68.9	33.1	85.6	31.4	84.6	30.1	82.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.2	25.1	22.1	-30.3	100.3	8.1	3.1	-36.1	102.5	-40.9	104.3	-45.4	106.1	-51.2	108.4		
			-28.2	71.6	6.1	3.9	-32.5	74.4	-36.1	76.6	-39.5	78.8	-43.9	81.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	17.7	20.5	31.5	43.8	53.5	61.8	70.4	68.5	57.9	44.3	30.9	20.8		
	DBStd	21.52	16.58	16.03	14.27	9.03	7.81	6.43	6.00	6.54	8.12	9.35	13.23	15.12		
	HDD50	4411	1003	828	579	221	51	3	0	0	23	220	577	906		
	HDD65	8219	1468	1247	1039	637	363	134	19	41	236	642	1022	1371		
	CDD50	2072	0	1	5	34	160	357	631	573	262	44	5	0		
	CDD65	404	0	0	0	0	7	38	185	149	24	1	0	0		
	CDH74	7105	0	0	1	35	261	740	2951	2424	658	34	1	0		
	CDH80	3189	0	0	0	5	65	242	1454	1163	257	3	0	0		
	WSAvg	10.0	10.3	9.8	10.4	10.9	10.5	9.9	9.0	8.9	9.1	10.0	10.4	10.6		
	PrecAvg	12.4	0.5	0.3	0.6	1.0	1.9	2.4	1.6	1.3	1.2	0.7	0.4	0.5		
	PrecMax	21.3	2.2	0.9	2.0	2.9	6.1	6.7	7.3	4.6	6.4	2.5	1.2	2.1		
Precipitation	PrecMin	6.8	0.0	0.0	0.0	0.0	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.8	0.4	0.2	0.5	0.7	1.5	1.5	1.5	1.2	1.1	0.5	0.4	0.4		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	56.9	68.6	78.8	86.6	93.1	99.6	99.1	92.6	79.2	66.1	53.7		
		MCWB	42.2	43.7	49.3	54.1	59.5	64.2	65.6	64.2	61.3	55.9	49.1	42.1		
	2%	DB	47.3	49.7	61.7	72.2	80.9	86.4	95.8	94.4	87.5	71.6	58.9	48.6		
		MCWB	39.0	39.6	46.1	52.1	58.1	62.3	64.1	62.8	59.9	52.3	45.5	38.9		
	5%	DB	42.7	44.7	56.1	66.7	76.0	81.9	92.5	90.9	82.0	65.8	53.3	44.4		
		MCWB	36.3	36.9	43.0	49.0	55.5	60.9	63.8	62.2	58.1	50.0	42.5	36.6		
	10%	DB	39.1	39.9	50.6	61.4	71.1	78.0	88.5	87.2	76.2	60.8	48.5	40.1		
		MCWB	34.4	34.3	40.5	46.3	53.0	59.2	62.9	61.3	55.8	47.4	39.9	34.1		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	44.5	50.6	56.1	61.5	68.2	71.9	68.5	65.4	57.5	51.1	43.6		
		MCDB	52.0	55.2	67.1	73.6	80.9	84.0	89.8	90.4	84.5	75.7	65.4	52.6		
	2%	WB	39.5	40.3	46.9	52.7	59.0	65.0	68.3	65.7	61.7	53.5	46.2	39.5		
		MCDB	46.5	48.6	59.4	70.6	77.2	81.8	86.5	87.3	80.9	68.5	57.5	47.3		
	5%	WB	36.5	37.0	43.8	50.0	56.9	62.7	66.1	64.0	59.2	51.0	43.1	36.8		
		MCDB	42.1	43.8	55.3	65.4	72.8	77.5	85.0	84.8	78.1	64.1	52.1	43.8		
	10%	WB	34.6	34.6	40.9	47.2	54.6	60.5	64.3	62.4	56.9	48.2	40.2	34.4		
		MCDB	39.2	39.9	50.1	59.9	68.2	74.3	83.5	82.6	74.8	59.7	48.0	40.0		
Mean Daily Temperature Range	5% DB	MDBR	19.5	20.4	22.3	25.3	25.7	25.8	31.1	31.2	28.9	25.3	21.1	19.8		
		MCDBR	21.7	24.8	31.6	35.7	35.1	33.4	37.5	38.6	40.0	34.5	27.3	23.2		
		MCWBR	15.3	16.5	18.0	18.5	16.0	13.9	13.4	14.7	17.6	18.4	16.3	15.5		
	5% WB	MCDBR	20.9	24.1	30.0	33.6	31.0	29.6	32.2	33.3	36.2	32.5	25.9	22.7		
		MCWBR	15.4	16.7	17.9	17.8	14.8	13.4	13.7	13.3	16.8	17.8	16.1	15.8		
Clear-Sky Solar Irradiance	taub		0.264	0.269	0.290	0.334	0.355	0.350	0.355	0.376	0.329	0.297	0.268	0.257		
	taud		2.452	2.494	2.489	2.371	2.356	2.423	2.414	2.330	2.431	2.517	2.509	2.448		
	Ebn at Noon		262	288	297	291	287	289	285	273	278	272	259	249		
	Edh at Noon		20	24	28	35	37	35	35	36	30	23	18	17		
All-Sky Solar Radiation	RadAvg		433	741	1144	1558	1865	2086	2245	1845	1328	841	490	336		
	RadStd		30	59	69	77	141	118	123	114	102	68	38	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page